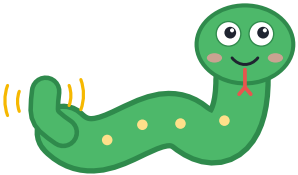
Recitation 1

Part 1: Intro and Python Basics

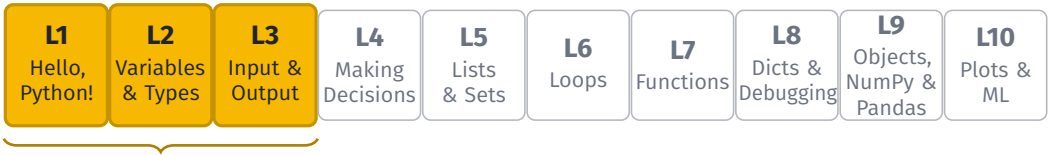
Recitation 1: Be the Computer

Read code like the intern, fill in the blanks, write programs that talk and decide



Dror Jacoby
Tel Aviv University

Where we are


L1
Hello, Python!
L2
Variables & Types
L3
Input & Output
L4
Making Decisions
L5
Lists & Sets
L6
Loops
L7
Functions
L8
Dicts & Debugging
L9
Objects, NumPy & Pandas
L10
Plots & ML
Today we practise L1–L3

Today's plan

◎ By the end of today you can…

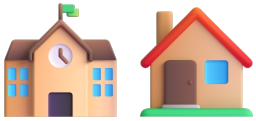
- Read code line by line, like the computer does
- Fill in missing pieces of a program
- Write programs with math, f-strings and if



Laptops open, Colab ready. Today **you** type. 💻

How recitations work

in class	Recap, warm-up, worked problems, and the exercises marked **IN CLASS**
at home	Exercises marked **AT HOME**: practice before the homework
solutions	At the end of this deck. Try first, peek later!



There is more here than one session. That is on purpose.

Topic 1

Recap and warm-up

Recitation 1 • Print, variables, input & if



TODAY

1. **Recap and warm-up**
2. Worked problems
3. Your turn
4. Exam corner
5. Solutions

Lecture 1 in one slide

Write	Python does	Shows
`print("2 + 3")`	copies the text	`2 + 3`
`print(2 + 3)`	calculates	`5`
`# a note`	nothing (comment)	
`Print("hi")`	NameError	capital P!
`print("hi)`	SyntaxError	missing quote



The intern runs lines top to bottom, exactly.

Lecture 2 in one slide: boxes and kinds

```
city = "Haifa"       # str
students = 32        # int
price = 7.5          # float
is_open = True       # bool
students = students + 1
print(students)
```

OUTPUT

```
33
```

- = means **store**, right side first
- A box holds **one** value at a time
- `type(x)` tells you the kind

Lecture 2 in one slide: the math symbols

Symbol	Meaning	Example
`/`	divide (always float)	`7 / 2` → 3.5
`//`	whole part	`7 // 2` → 3
`%`	leftover	`7 % 2` → 1
`**`	power	`2 ** 3` → 8
`"ab" * 2`	repeat text	abab
`"a" + "b"`	glue text	ab



`//` and `%` are a pair:
“how many fit” and “what’s left”.

Lecture 3 in one slide: talk and decide

```
name = input("Your name? ")
age = int(input("Your age? "))
print(f"Hi {name}! Next year: {age + 1}")
price = 7 / 3
print(f"{price:.2f}")
```

OUTPUT

```
Your name? Dana
Your age? 20
Hi Dana! Next year: 21
2.33
```

- `input()` always gives **text**
- `int()`, `float()` turn it into a number
- `f"...{x}..."` fills in values
- `:.2f` = two digits after the dot
- **`if`**/**`else`** picks one of two roads

Warm-up 1: text times a number

Predict the output!

```
print("7" * 3)
print(7 * 3)
```



Warm-up 1: text times a number

Predict the output!

```
print("7" * 3)
print(7 * 3)
```

Answer

777
21

Quotes: repeat the text. No quotes: real math.



Warm-up 2: follow the box

Predict the output!

```
likes = 10
likes = likes + 5
likes = likes * 2
print(likes)
```

Warm-up 2: follow the box

Predict the output!

```
likes = 10
likes = likes + 5
likes = likes * 2
print(likes)
```

Answer

`30`

10, then 15, then 30. Each line uses the *current* value.

Warm-up 3: whole part, leftover, power

Predict the output!

```
print(17 // 5)
print(17 % 5)
print(2 ** 5)
```

Warm-up 3: whole part, leftover, power

Predict the output!

```
print(17 // 5)
print(17 % 5)
print(2 ** 5)
```

Answer

3
2
32

5 fits 3 times in 17, 2 left over.
$2 \times 2 \times 2 \times 2 \times 2 = 32$.

Warm-up 4: tiny traps

Predict the output!

```
print(10 / 2)
print(type(10 / 2))
print(len("Tel Aviv"))
```

Warm-up 4: tiny traps

Predict the output!

```
print(10 / 2)
print(type(10 / 2))
print(len("Tel Aviv"))
```

Answer

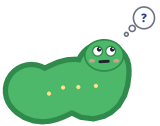
```
5.0
<class 'float'>
8
```

`/` always gives a float. The space is a character too!

Warm-up 5: text or number?

Predict the output!

```
age = "20"
print(age + "1")
print(int(age) + 1)
print(f"{7 / 3:.2f}")
```



Warm-up 5: text or number?

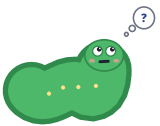
Predict the output!

```
age = "20"
print(age + "1")
print(int(age) + 1)
print(f"{7 / 3:.2f}")
```

Answer

```
201
21
2.33
```

Text + text glues. `int()` makes it a number first.



Topic 2

Worked problems

Recitation 1 • Print, variables, input & if



TODAY

1. Recap and warm-up
2. **Worked problems**
3. Your turn
4. Exam corner
5. Solutions

Problem 1: the falafel receipt

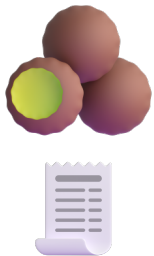
Pip's Falafel needs a receipt

3 falafels at 14 ILS, 2 drinks at 9 ILS.
Print a title, the falafel cost, and the total.

Problem 1: the falafel receipt

Pip's Falafel needs a receipt

3 falafels at 14 ILS, 2 drinks at 9 ILS.
Print a title, the falafel cost, and the total.



Plan in words first:

1. Print a title
2. Print the falafel line and its cost
3. Print the total

Build it, one step at a time

```
# Plan: title, items, total
print("== Pip's Falafel ==")
```

Build it, one step at a time

```
# Plan: title, items, total
print("== Pip's Falafel ==")
print("Falafel x3")
print(3 * 14)
```

Build it, one step at a time

```
# Plan: title, items, total
print("== Pip's Falafel ==")
print("Falafel x3")
print(3 * 14)
print("Total in ILS:")
print(3 * 14 + 2 * 9)
```

Build it, one step at a time

```
# Plan: title, items, total
print("== Pip's Falafel ==")
print("Falafel x3")
print(3 * 14)
print("Total in ILS:")
print(3 * 14 + 2 * 9)
```

OUTPUT

```
== Pip's Falafel ==
Falafel x3
42
Total in ILS:
60
```

Build it, one step at a time

```
# Plan: title, items, total
print("== Pip's Falafel ==")
print("Falafel x3")
print(3 * 14)
print("Total in ILS:")
print(3 * 14 + 2 * 9)
```

OUTPUT

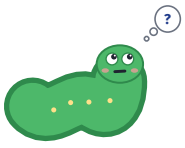
```
== Pip's Falafel ==
Falafel x3
42
Total in ILS:
60
```

The comments are the plan. The code follows the plan, line by line.

What if...?

Discuss with your neighbour

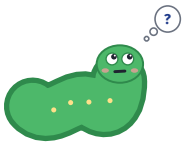
1. What does `print("3 * 14")` show?
2. A falafel now costs 16 ILS. How many places change?



What if...?

Discuss with your neighbour

1. What does `print("3 * 14")` show?
2. A falafel now costs 16 ILS. How many places change?



Answer

1. The text `3 * 14`, not 42.
2. Two places! That is why we need **variables**.

Problem 2: how many bus rides?

Your Rav-Kav

You have 50 ILS on the card. A ride costs 5.5 ILS.
How many rides can you take? What is left?

Problem 2: how many bus rides?

Your Rav-Kav

You have 50 ILS on the card. A ride costs 5.5 ILS.
How many rides can you take? What is left?



Plan in words:

1. Store the balance and the ride price
2. Rides = how many times the price *fits* `//`
3. Left = what is *left over* `%`

Code it

rav_kav.py

```
balance = 50
ride = 5.5
rides = balance // ride
left = balance % ride
print(rides)
print(left)
```

Code it

rav_kav.py

```
balance = 50
ride = 5.5
rides = balance // ride
left = balance % ride
print(rides)
print(left)
```

OUTPUT

```
9.0
0.5
```

Code it

rav_kav.py

```
balance = 50
ride = 5.5
rides = balance // ride
left = balance % ride
print(rides)
print(left)
```

OUTPUT

```
9.0
0.5
```

Why 9.0 and not 9?

One float in the math (5.5) makes the answer a float. Still 9 rides!

Joke break: the mathematician and the kettle

Task 1: an **empty** kettle. Make tea.
Both: fill the kettle, turn it on. ✓

Joke break: the mathematician and the kettle

Task 1: an **empty** kettle. Make tea.
Both: fill the kettle, turn it on. ✓

Task 2: the kettle is **already full**. Make tea.
Physicist: just turns it on.

Joke break: the mathematician and the kettle

Task 1: an **empty** kettle. Make tea.
Both: fill the kettle, turn it on. ✓

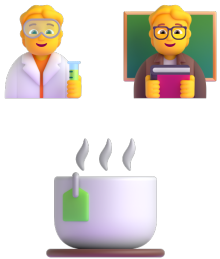
Task 2: the kettle is **already full**. Make tea.
Physicist: just turns it on.
Mathematician: **pours the water out**.
“Now it’s Task 1, and I already solved that.”

Joke break: the mathematician and the kettle

Task 1: an **empty** kettle. Make tea.
Both: fill the kettle, turn it on. ✓

Task 2: the kettle is **already full**. Make tea.
Physicist: just turns it on.
Mathematician: **pours the water out**.
“Now it’s Task 1, and I already solved that.”

Programmers do it too: turn a new problem into one you already solved, and **reuse** that plan. Next: same program, new numbers.



What if you top up to 100 ILS?

Predict the output!

```
balance = 100
ride = 5.5
print(balance // ride)
print(balance % ride)
```

What if you top up to 100 ILS?

Predict the output!

```
balance = 100
ride = 5.5
print(balance // ride)
print(balance % ride)
```

Answer

```
18.0
1.0
```

We changed **one** line.
Variables make updates easy.

Level up: seconds to hours, minutes, seconds

```
seconds = 10000
hours = seconds // 3600
rest = seconds % 3600
minutes = rest // 60
secs = rest % 60
print(hours)
print(minutes)
print(secs)
```

Level up: seconds to hours, minutes, seconds

```
seconds = 10000
hours = seconds // 3600
rest = seconds % 3600
minutes = rest // 60
secs = rest % 60
print(hours)
print(minutes)
print(secs)
```

OUTPUT

```
2
46
40
```

Level up: seconds to hours, minutes, seconds

```
seconds = 10000
hours = seconds // 3600
rest = seconds % 3600
minutes = rest // 60
secs = rest % 60
print(hours)
print(minutes)
print(secs)
```

OUTPUT

```
2
46
40
```

10,000 seconds = 2 h 46 min 40 s.

Same pair, used twice: `//` then `%`.

Problem 3: will my Rav-Kav last the week?

Decide

The balance is in a variable. 10 rides cost 5.5 ILS each. Enough? Say what is left. If not, say how much to top up.



Problem 3: will my Rav-Kav last the week?

Decide

The balance is in a variable. 10 rides cost 5.5 ILS each. Enough? Say what is left. If not, say how much to top up.



Plan in words:

1. Store the balance in a variable
2. Cost = 10 rides $\times$ 5.5
3. **If** balance $\geq$ cost: print what is left
4. **Else**: print how much to top up

Code it, run it twice

```
balance = 50   # then try 80
rides = 10
cost = rides * 5.5
if balance >= cost:
    print(f"OK! {balance - cost:.2f} ILS left")
else:
    print(f"Top up {cost - balance:.2f} ILS")
```

Code it, run it twice

```
balance = 50   # then try 80
rides = 10
cost = rides * 5.5
if balance >= cost:
    print(f"OK! {balance - cost:.2f} ILS left")
else:
    print(f"Top up {cost - balance:.2f} ILS")
```

Run 1: `balance = 50`

OUTPUT

```
Top up 5.00 ILS
```

Code it, run it twice

```python
balance = 50   # then try 80
rides = 10
cost = rides * 5.5
if balance >= cost:
    print(f"OK! {balance - cost:.2f} ILS left")
else:
    print(f"Top up {cost - balance:.2f} ILS")
```

Run 1: `balance = 50`

OUTPUT

```
Top up 5.00 ILS
```

Run 2: `balance = 80`

OUTPUT

```
OK! 25.00 ILS left
```

Code it, run it twice

```python
balance = 50   # then try 80
rides = 10
cost = rides * 5.5
if balance >= cost:
    print(f"OK! {balance - cost:.2f} ILS left")
else:
    print(f"Top up {cost - balance:.2f} ILS")
```

Run 1: `balance = 50`

OUTPUT

```
Top up 5.00 ILS
```

Run 2: `balance = 80`

OUTPUT

```
OK! 25.00 ILS left
```

Test both roads of the `if`: change the value, run again. (With `input()` the user types it.)

Topic 3

Your turn

Recitation 1 • Print, variables, input & if



TODAY

1. Recap and warm-up
2. Worked problems
3. **Your turn**
4. Exam corner
5. Solutions

Your turn: 8 exercises

Exercise	Level	Where
1 Fix the welcome screen	★	in class
2 The sale banner	★	in class
3 Change in coins	★★	in class
4 Road-trip playlist	★★	at home
5 Pizza party planner	★★★	at home
6 Coffee receipt (f-strings)	★	in class
7 Pass or fail	★★	at home
8 Shekels to dollars	★★★	at home



Predict the output
before you run.
Stuck for 3 minutes? Ask!

Exercise 1 ⭐: fix the welcome screen

EXERCISE 1

This code has 3 bugs. Fix them all.

```
Print("Welcome to TAU")
print("Room 110)
print "Have fun"
```

Wanted output:

OUTPUT

```
Welcome to TAU
Room 110
Have fun
```

Exercise 1 ⭐: fix the welcome screen

EXERCISE 1

This code has 3 bugs. Fix them all.

```
Print("Welcome to TAU")
print("Room 110)
print "Have fun"
```

Wanted output:

OUTPUT

```
Welcome to TAU
Room 110
Have fun
```

Hint

Fix one line, run, read the **last line** of the error, repeat.

Exercise 2 ⭐: the sale banner

EXERCISE 2

Fill in the yellow blanks

```
print("=" * ____)
print("  SALE 50%")
print("=" * ____)
```

Wanted output:

OUTPUT

```
============
  SALE 50%
============
```

Exercise 2 ⭐: the sale banner

EXERCISE 2

Fill in the yellow blanks

```
print("=" * ____)
print("  SALE 50%")
print("=" * ____)
```

Wanted output:

OUTPUT

```
============
  SALE 50%
============
```

Hint

Count the = signs in one line of the wanted output.

Exercise 3 ⭐⭐: change in coins

EXERCISE 3

37 ILS change in 10s, 5s and 1s: fill in

```
change = 37
tens = change ___ 10
rest = change ___ 10
fives = rest ___ 5
ones = rest ___ 5
print(tens)
print(fives)
print(ones)
```

Wanted output: 🪙

OUTPUT

```
3
1
2
```

Exercise 3 ⭐⭐: change in coins

EXERCISE 3

37 ILS change in 10s, 5s and 1s: fill in

```
change = 37
tens = change  ___  10
rest = change  ___  10
fives = rest   ___  5
ones = rest    ___  5
print(tens)
print(fives)
print(ones)
```

Wanted output: 🪙

OUTPUT

```
3
1
2
```

Hint

"How many fit" or "what is left"? Each blank is `//` or `%`.

Exercise 4 ⭐⭐: the road-trip playlist

AT HOME 4



Write it from scratch

Your playlist: **40** songs, about **3.5** minutes each. Print:

1. the total minutes
2. the total in hours (with a fraction)
3. whole hours, then leftover minutes

Exercise 4 ⭐⭐: the road-trip playlist

AT HOME 4

Write it from scratch

Your playlist: **40** songs, about **3.5** minutes each. Print:

1. the total minutes
2. the total in hours (with a fraction)
3. whole hours, then leftover minutes



Hint

Use 3 variables: `songs`, `song_minutes`, `total_minutes`.

Exercise 5 ⭐⭐⭐: the pizza party planner

AT HOME 5

23 students, 3 slices each, pizza = 8 slices, 52 ILS

```
students = 23
slices_each = 3
slices_needed = 
pizzas = (slices_needed + 7) // 8
cost = 
print(slices_needed)
print(pizzas)
print(cost)
print(  )  # per student
```

Wanted output: 🍕

OUTPUT

```
69
9
468
20.347826086956523
```

Exercise 5 ⭐⭐⭐: the pizza party planner

AT HOME 5

23 students, 3 slices each, pizza = 8 slices, 52 ILS

```
students = 23
slices_each = 3
slices_needed = 
pizzas = (slices_needed + 7) // 8
cost = 
print(slices_needed)
print(pizzas)
print(cost)
print(                )  # per student
```

Wanted output: 🍕

OUTPUT

```
69
9
468
20.347826086956523
```

Why + 7?

69 slices = 8.6 pizzas. “+ 7” then // rounds **up** to 9.

Exercise 6 ⭐: the coffee receipt

EXERCISE 6

fill in the f-string

```
name = "Noa"
price = 14.5
cups = 2
total = price * cups
print(f"      , you pay       ILS")
```

Wanted output: ☕

OUTPUT

```
Noa, you pay 29.00 ILS
```

Exercise 6 ⭐: the coffee receipt

EXERCISE 6

fill in the f-string

```
name = "Noa"
price = 14.5
cups = 2
total = price * cups
print(f"      , you pay       ILS")
```

Wanted output: 

OUTPUT

```
Noa, you pay 29.00 ILS
```

Hint

Put variable names inside `{ }`. Two digits: `:.2f`.

Exercise 7 ⭐⭐: pass or fail

AT HOME 7

Write it from scratch

Start with `grade = 72`. If it is 56 or more, print
`Passed! Well done`, otherwise
`Not this time. Try again!`



Two test runs:

`grade = 72` → `Passed! Well done`

`grade = 41` → `Not this time. Try again!`

Exercise 7 ⭐⭐: pass or fail

AT HOME 7

Write it from scratch

Start with `grade = 72`. If it is 56 or more, print
`Passed! Well done`, otherwise
`Not this time. Try again!`

Two test runs:

`grade = 72` → `Passed! Well done`

`grade = 41` → `Not this time. Try again!`



Hint

Change the value, run again: one run per road of the `if`.

Exercise 8 ⭐⭐⭐: shekels to dollars

AT HOME 8

🖉 Write it from scratch

Start with `shekels = 500`. 💵 Use an example rate of **3.7** shekels per dollar. Small changes (under 100 ILS) pay a **5 ILS fee** first. Print the dollars with 2 digits.

Two test runs:

```
shekels = 500
You get 135.14 dollars
shekels = 50
You get 12.16 dollars
```

3.7 is an example rate. Real rates change every day.

Exercise 8 ⭐⭐⭐: shekels to dollars

AT HOME 8

Write it from scratch

Start with `shekels = 500`. 💵 Use an example rate of **3.7** shekels per dollar. Small changes (under 100 ILS) pay a **5 ILS fee** first. Print the dollars with 2 digits.

Two test runs:

```
shekels = 500
You get 135.14 dollars
shekels = 50
You get 12.16 dollars
```

Hint

An `if` without `else` is fine: only small amounts lose the fee.

3.7 is an example rate. Real rates change every day.

Your home practice list

Before the next recitation

- Exercises 4, 5, 7 and 8 **at home**
- More `input()` practice: `L03_05_self_exercise`
- Any in-class exercise you did not finish
- Then check yourself against the solutions



Type the code yourself. Reading a solution is not practice.

Topic 4

Exam corner

Recitation 1 • Print, variables, input & if



TODAY

1 Recap and warm-up

2 Worked problems

3 Your turn

4 Exam corner

5 Solutions

Exam style 1: trace the code

What is printed? Write every line.

```
a = 7
b = a // 2
a = a % 2
c = b * 10 + a
print(c)
print(type(c))
```

Exam style 1: trace the code

What is printed? Write every line.

```
a = 7
b = a // 2
a = a % 2
c = b * 10 + a
print(c)
print(type(c))
```

Answer

31
<class 'int'>

Trace table: a=7, b=3, then a=1, c=$3 \times 10 + 1$=31.

Exam tip: keep a small table of every box and update it line by line.

Exam style 2: no error, wrong answer

Should print 80. Find the bug.

```
average = 80 + 90 + 70 / 3
print(average)
```

OUTPUT

```
193.33333333333334
```

Exam style 2: no error, wrong answer

Should print 80. Find the bug.

```
average = 80 + 90 + 70 / 3
print(average)
```

OUTPUT

```
193.33333333333334
```

Fix: parentheses

```
average = (80 + 90 + 70) / 3
print(average)
```

OUTPUT

```
80.0
```

/ goes before +. The intern did exactly what we wrote.

Three new mistakes to avoid

```
5 = x
```

ERROR

```
SyntaxError: cannot assign to literal here. Maybe you meant '==' instead of '='?
```

The box name goes on the **left**: `x = 5`

```
print("10" * "2")
```

ERROR

```
TypeError: can't multiply sequence by non-int of type 'str'
```

Text can be repeated by a **number**, not by text.

```
price = 12
print("price")
print(price)
```

OUTPUT

```
price
12
```

No error, but quotes print the **name**, not the value.

Before the homework

⚠ HW1: Introduction

All of Lectures 1–3, `if` / `else` too: one part per lecture, then bigger problems that mix them. A Colab notebook, submitted on Moodle by ∼**Nov 20**.



- Write the plan as comments first
- Run after every 2-3 lines, not at the end

Recap

✓ Today you learned

- Plan in words, then code it line by line
- `//` and `%` answer “how many fit” and “what’s left”
- `input()` gives text: convert it, then decide with `if`



Topic 5

Solutions

Recitation 1 • Print, variables, input & if



TODAY

1. Recap and warm-up
2. Worked problems
3. Your turn
4. Exam corner
5. **Solutions**

Solution 1: fix the welcome screen

SOLUTION

```
print("Welcome to TAU")
print("Room 110")
print("Have fun")
```

OUTPUT

```
Welcome to TAU
Room 110
Have fun
```

- Line 1: `print` with a small p
- Line 2: close the quote
- Line 3: add the parentheses

Solutions 2 and 3: banner and coins

SOLUTION

Exercise 2

```
print("=" * 12)
print("  SALE 50%")
print("=" * 12)
```

Count once, use the number twice.

Exercise 3

```
change = 37
tens = change // 10
rest = change % 10
fives = rest // 5
ones = rest % 5
```

`//` for “how many”, `%` for “what’s left”: 3, 1, 2.

Solution 4: the road-trip playlist

SOLUTION

```
songs = 40
song_minutes = 3.5
total_minutes = songs * song_minutes
print(total_minutes)
print(total_minutes / 60)
print(total_minutes // 60)
print(total_minutes % 60)
```

OUTPUT

```
140.0
2.3333333333333335
2.0
20.0
```

2 hours and 20 minutes of music.

Long decimals? Round them with `:.2f` from Lecture 3.

Solution 5: the pizza party planner

SOLUTION

```
students = 23
slices_each = 3
slices_needed = students * slices_each
pizzas = (slices_needed + 7) // 8
cost = pizzas * 52
print(slices_needed)
print(pizzas)
print(cost)
print(cost / students)  # per student
```

OUTPUT

```
69
9
468
20.347826086956523
```

About 20.35 ILS each. The “+ 7” trick made sure nobody stays hungry.

Solution 6: the coffee receipt

SOLUTION

```
name = "Noa"
price = 14.5
cups = 2
total = price * cups
print(f"{name}, you pay {total:.2f} ILS")
```

OUTPUT

```
Noa, you pay 29.00 ILS
```

`29.0` becomes `29.00`: money always shows two digits.

Solution 7: pass or fail

SOLUTION

```
grade = 72   # then try 41
if grade >= 56:
    print("Passed! Well done")
else:
    print("Not this time. Try again!")
```

OUTPUT

```
Passed! Well done
```

OUTPUT

```
Not this time. Try again!
```

Solution 8: shekels to dollars

SOLUTION

```
shekels = 500   # then try 50
rate = 3.7
dollars = shekels / rate
if shekels < 100:
    dollars = (shekels - 5) / rate
print(f"You get {dollars:.2f} dollars")
```

OUTPUT
```
You get 135.14 dollars
```

OUTPUT
```
You get 12.16 dollars
```

The `if` replaces the answer only for small amounts.

3D emoji: Microsoft Fluent Emoji (MIT License). Pip the snake: original drawing. Full list: images/CREDITS.md